Lecture 2: Linear Models for Regression

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SoE, Westlake University

February 24, 2025

WESTLAKE UNIVERSITY | SCHOOL OF ENGINEERING

1. Regression and Classification
   - Regression
   - Classification

2. Linear Regression
   - Definition of Linear Regression
   - Optimization Basics
   - Gradient Descent (GD) variants for Linear Regression
   - Normal Equations and Least Squares

This lecture:

- Basic concept of regression and classification
- Linear regression
  - Definition
  - Gradient Descent (GD) optimization
- Normal Equation and Least Squares (maybe?)

Next lecture:

- Normal Equation and Least Squares
- The probabilistic interpretation of Linear Regression
- Maximum Likelihood Estimation (MLE)
- Over-fitting and Under-fitting
- Polynomial Regression, Ridge Regression, and Lasso

Reading materials

- Chapter 1, Stanford CS 229 Lecture Notes, `https://cs229.stanford.edu/notes2022fall/main_notes.pdf`
- Chapter 3.1, Bishop, Pattern Recognition and Machine Learning

Reference

- EPFL, CS-433 Machine Learning, `https://github.com/epfml/ML_course`

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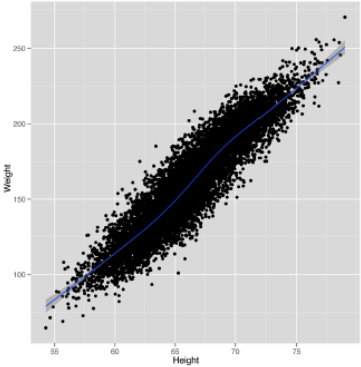
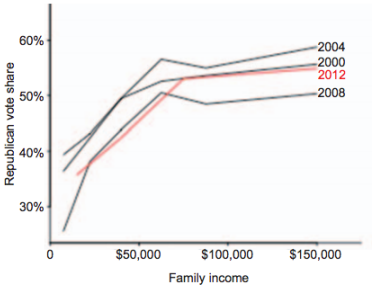
What is regression?



(a) Height is correlated with weight. Taken from “Machine Learning for Hackers"



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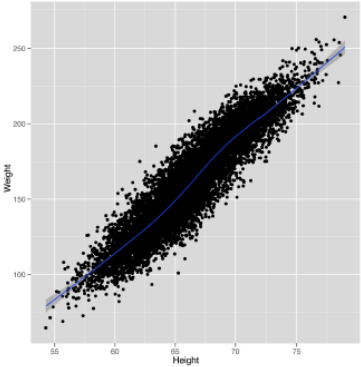

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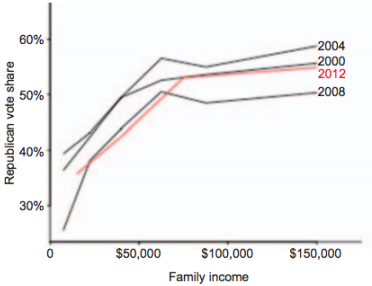

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Regression is to relate input variables to the output variable.

Dataset for regression

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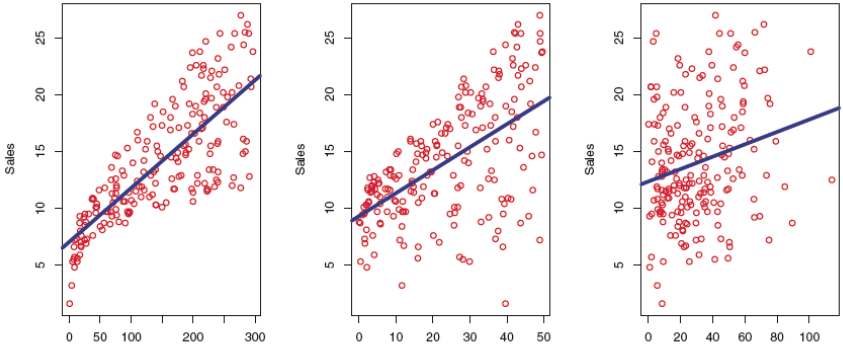

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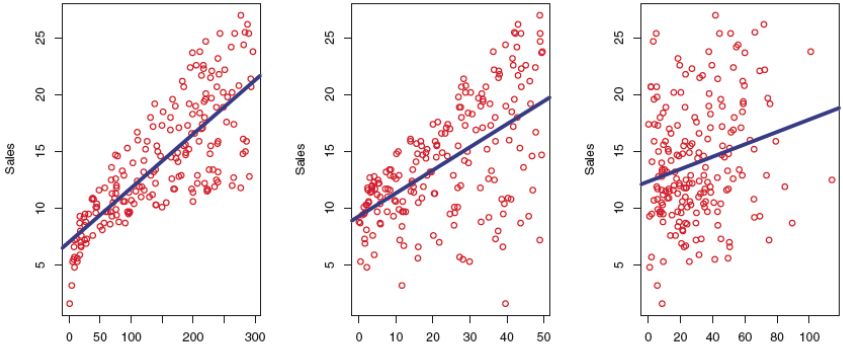

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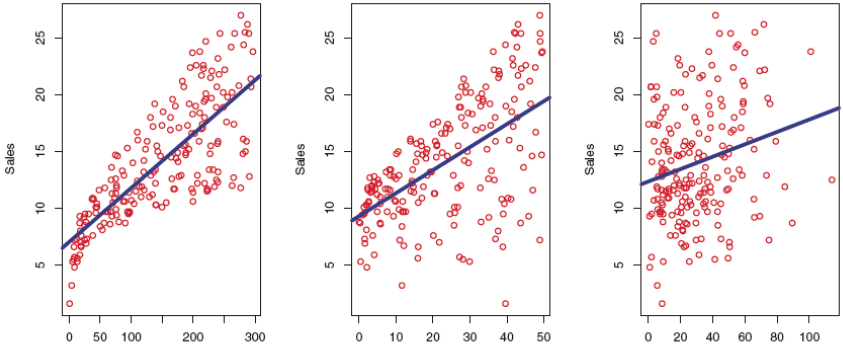
- The number of pairs N is the data-size and D is the dimensionality.



Two goals of regression

The regression function approximates the output y_n "well enough" given inputs $\mathbf{x}_n$.

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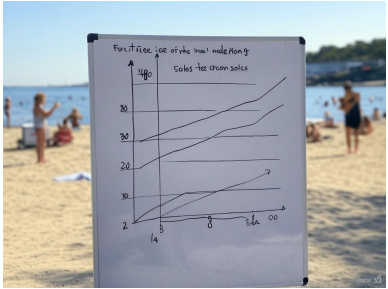
1. prediction: predict outputs for new inputs.

   e.g., what is the weight of a person who is 170 cm tall?

2. interpretation: understand the effect of the input on the output.

   e.g., are taller people heavier too?

Why correlation isn't causation: spurious correlation



ice cream sales vs. drowning incidents

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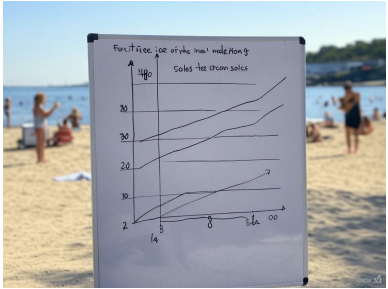

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Remark 1 (Correlation $\neq$ Causation)

Regression finds a correlation not a causal relationship, so interpret your results with caution.

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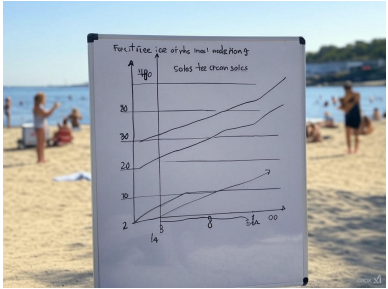

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Brainstorm other examples!

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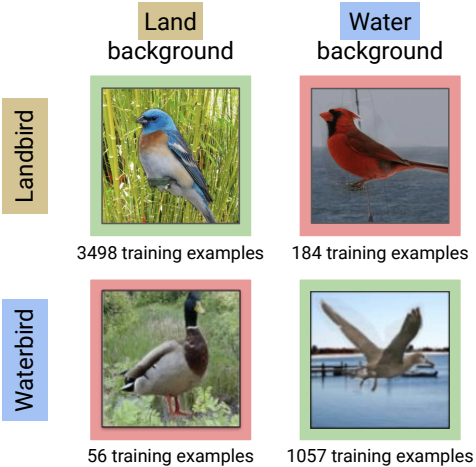
What is the concept of “spurious correlations”?



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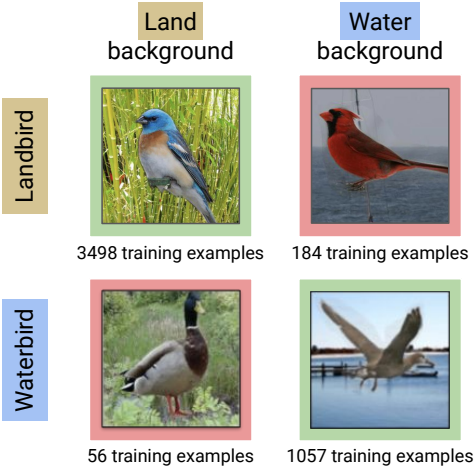
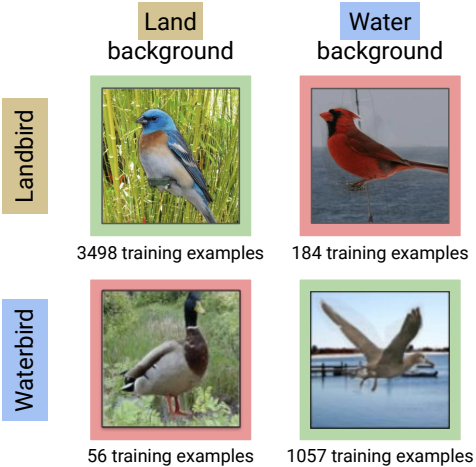


What is the concept of "spurious correlations"?

- The training set $\mathcal{D}_{tr}$ presents two spurious correlations, $\langle y, a \rangle$ and $\langle y', a \rangle$, i.e., a exists in two classes of images and can be mapped to two class labels.



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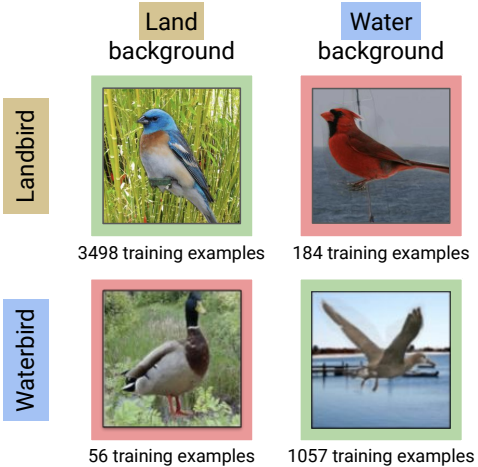
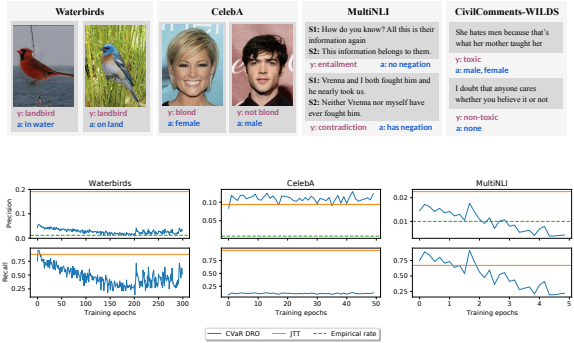
- A model trained on $\mathcal{D}_{tr}$ may simply use a to predict y, but this will result in incorrect predictions on samples with $\langle y', a \rangle$.

	Land background	Water background
Landbird	3498 training examples	184 training examples
Waterbird	56 training examples	1057 training examples

Remark 2 (Shortcut learning in Deep Learning)

Models may only learn spurious correlation (and thus sensitive to distribution shifts).





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A formal definition of spurious correlation

Definition 3 (Spurious Correlation)

Let $\mathcal{D}_{tr} = \{(x_i, y_i)\}_{i=1}^{n}$ be the training set with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, where $\mathcal{X}$ denotes the set of all possible inputs, $\mathcal{Y}$ denotes the set of K classes.

- For each data point x_i with class label y_i, there is a spurious attribute $a_i \in \mathcal{A}$ of x_i, where a_i is non-predictive of y_i, and $\mathcal{A}$ denotes the set of all possible spurious attributes.

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- A *spurious correlation*, denoted as $\langle y, a \rangle$, is the association between $y \in \mathcal{Y}$ and $a \in \mathcal{A}$, where y and a exist in a one-to-many mapping $\phi : \mathcal{A} \mapsto \mathcal{Y}^{K'}$ conditioned on $\mathcal{D}_{tr}$ with $1 < K' \leq K$.

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- A set of data with the spurious correlation $\langle y, a \rangle$ is annotated with the group label $g = (y, a) \in \mathcal{G}$, where $\mathcal{G} := \mathcal{Y} \times \mathcal{A}$ is the set of combinations of class labels and spurious attributes.

Wait!

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What is the definition of “classes”?

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We observe some data

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Remark 4

no ordering between classes.

Classifier

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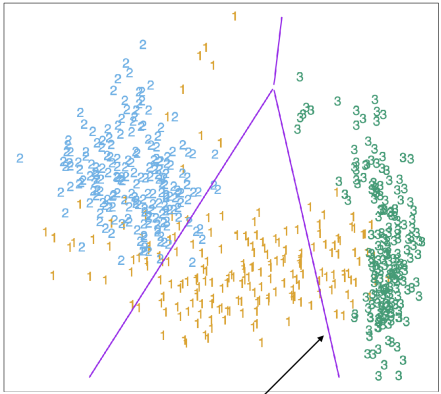

Linear Decision boundary

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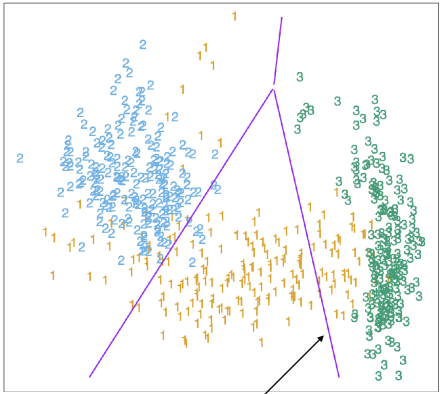

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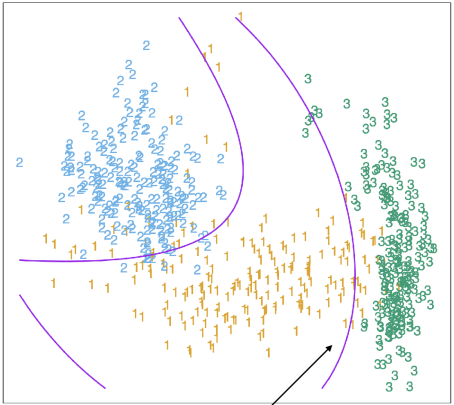

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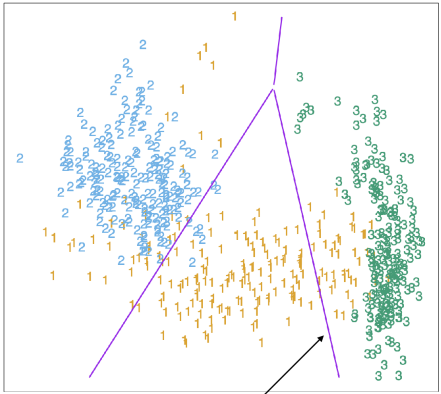

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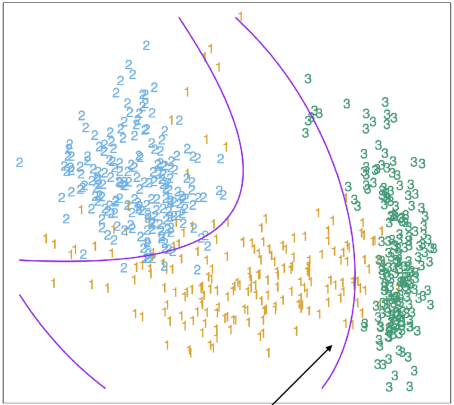

Nonlinear Decision boundary

The boundaries of these regions are called decision boundaries.

Classification: a special case of regression?

Classification is a **regression problem** with discrete labels:

$$(\mathbf{x}, y) \in \mathcal{X} \times \{0, 1\} \subset \mathcal{X} \times \mathbb{R} \tag{4}$$

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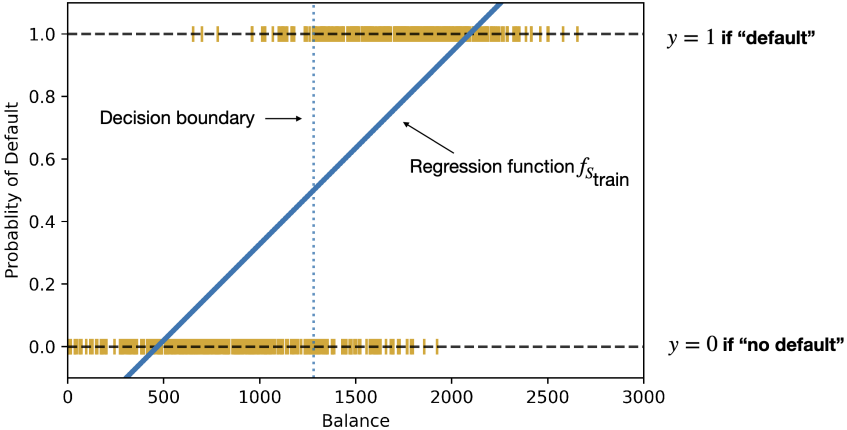
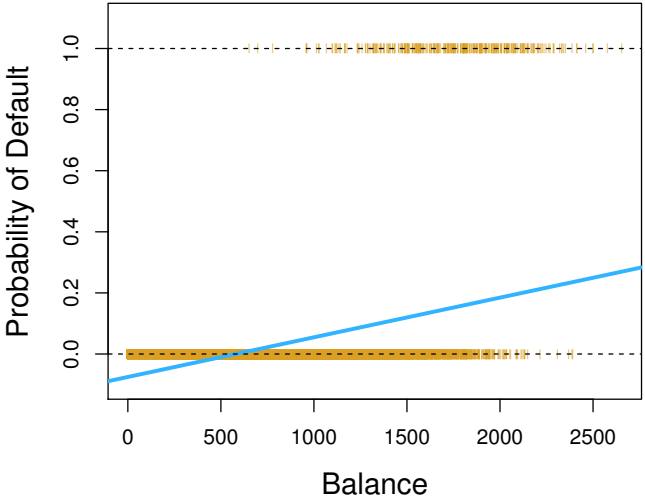
Could we use previously seen regression methods to solve it?

Is it a good idea to use regression methods for classification tasks?



We label the output as probability for sake of interpretation

Classification is not just a special form of regression



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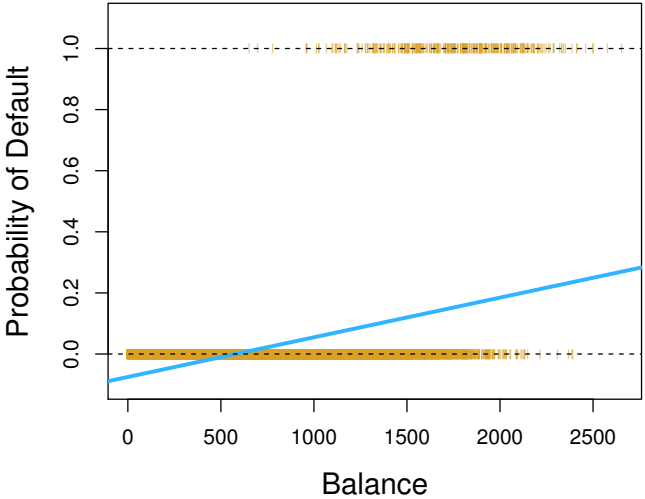
- The predicted values are not probabilities (not in $[0, 1]$).



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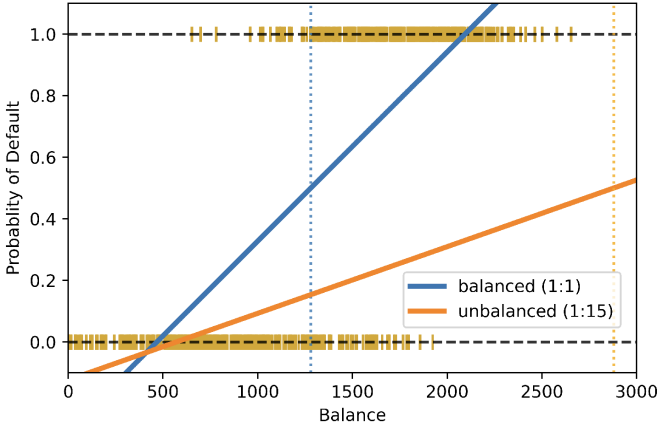

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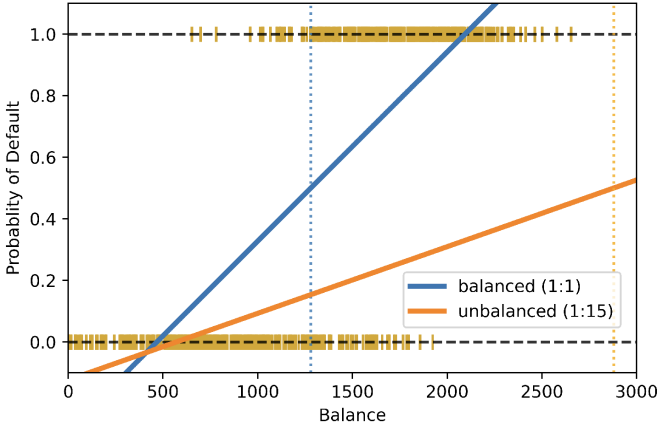
- Sensitivity to unbalanced data.



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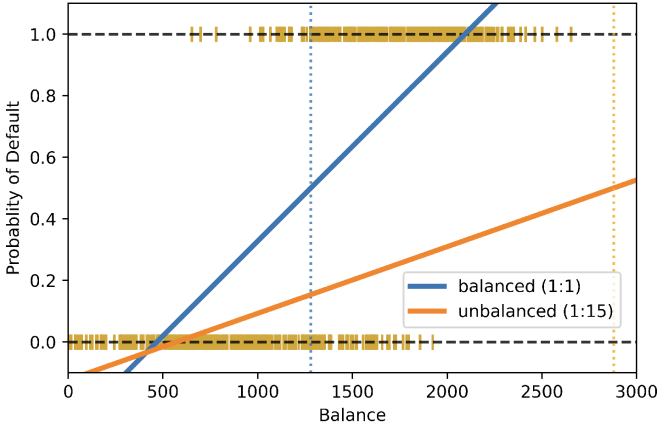
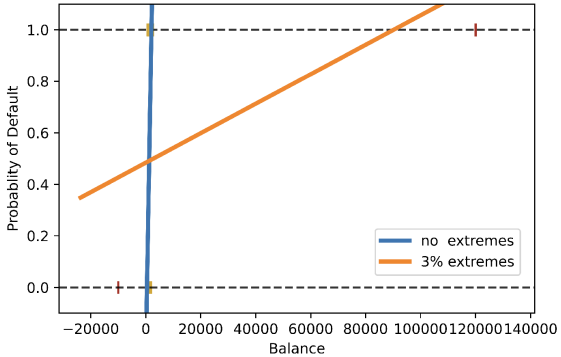
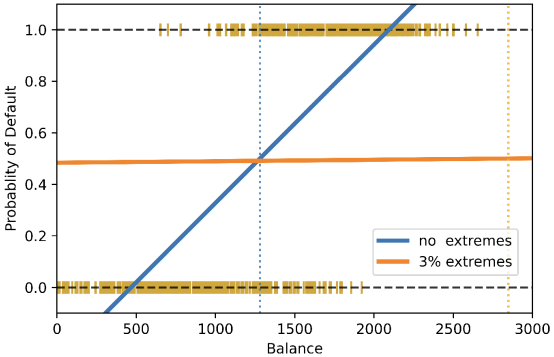


The position of the line depends crucially on how many points are in each class.

Classification is not just a special form of regression





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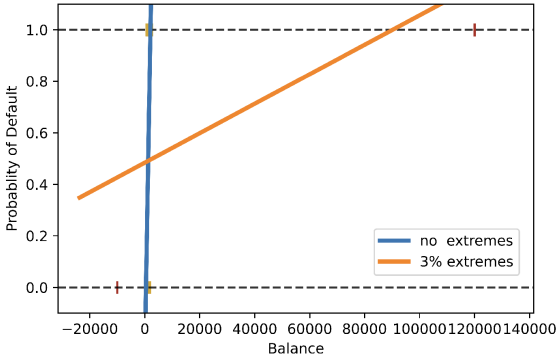
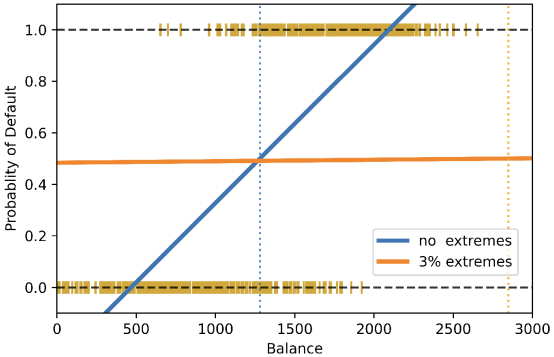
- Sensitivity to extreme values:





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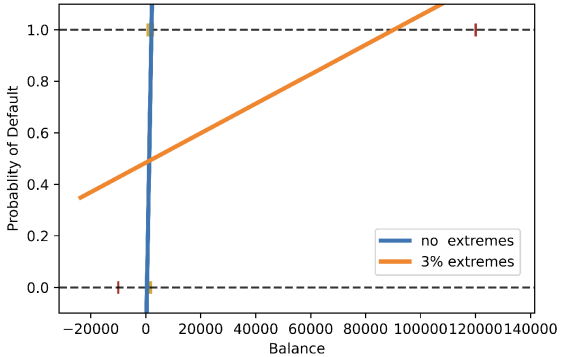

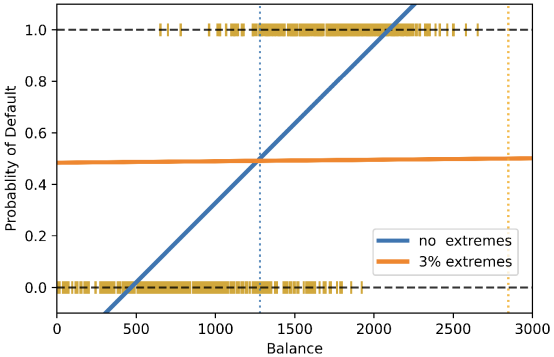

The position of the line depends crucially on where the points lie.

How to perform classification?

- A lot of different approaches have been developed

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- Fundamental task of classification:

separate the space into various decision regions

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Definition

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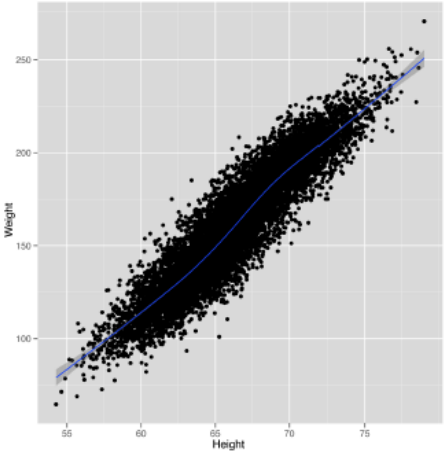

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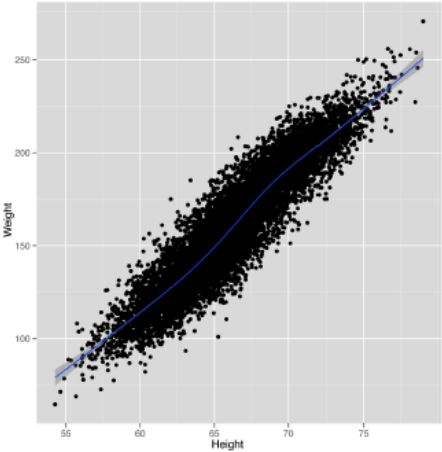

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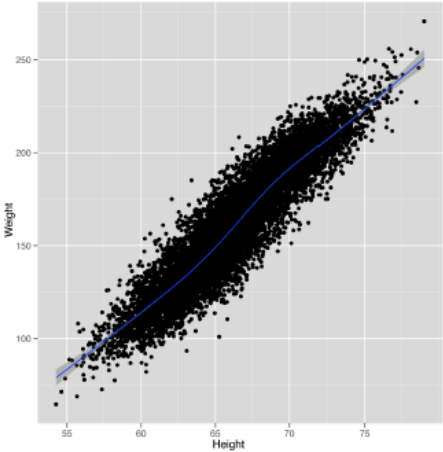

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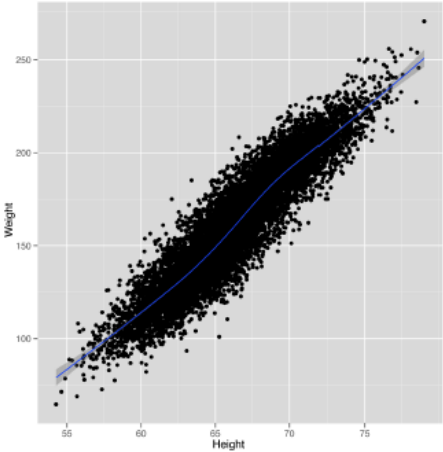

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We need an optimization algorithm!

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- we can learn almost all fundamental concepts of ML with regression alone

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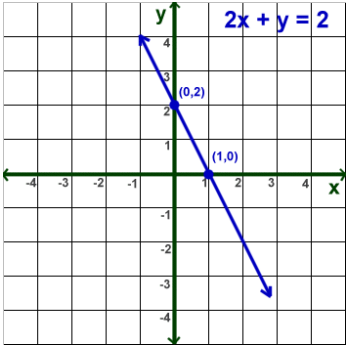
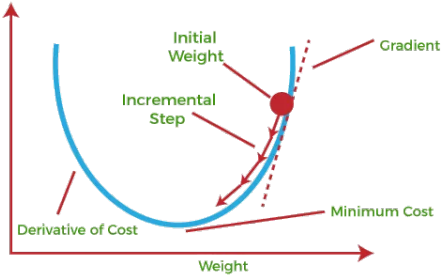
Q: How can we estimate values of $\mathbf{w}$ given the data $\mathcal{D}$?

Analytical vs. Numerical solutions

- ***Analytical solutions*** provide precise closed-form solutions but are limited to simple problems.

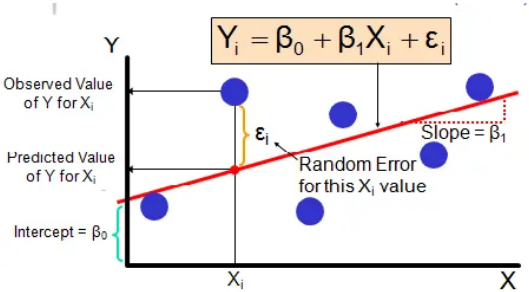


- ***Gradient Descent*** (GD) is a numerical method that approximates the optimal solution iteratively for complex problems.



https://medium.com/@msong507/gradient-descent-intuitive-overview-43e7869e1545

Approximate optimal solution $\mathbf{w}^\star$: cost function



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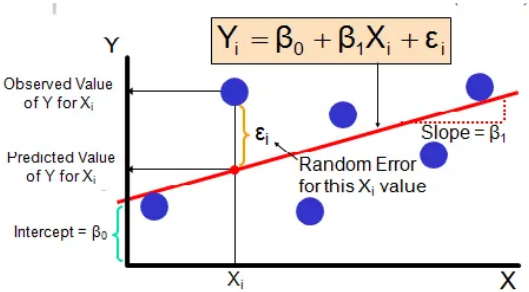

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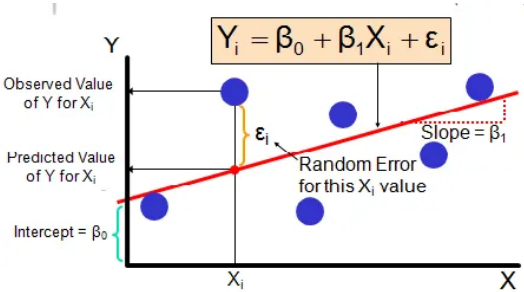

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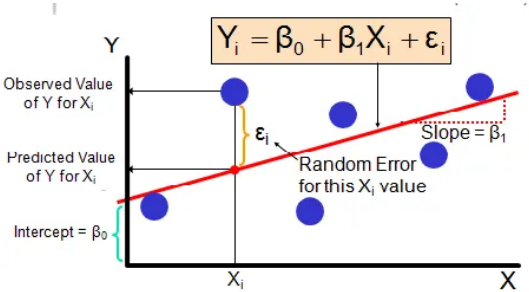

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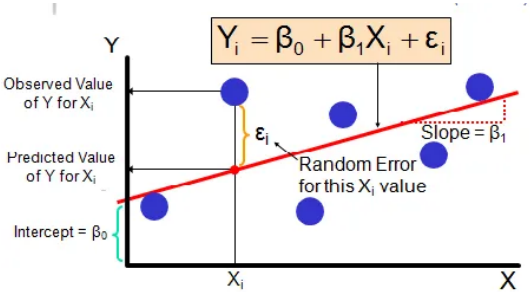

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Mean Squared Error (MSE) and outliers

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Discussion on different cost functions

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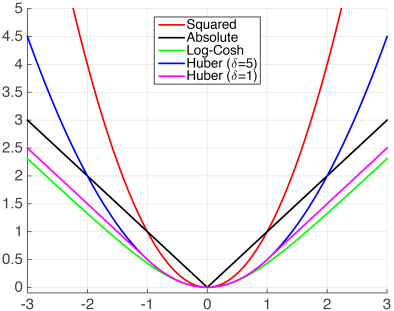

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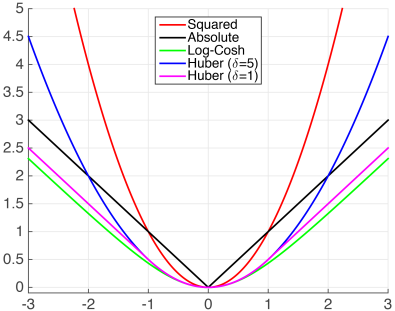
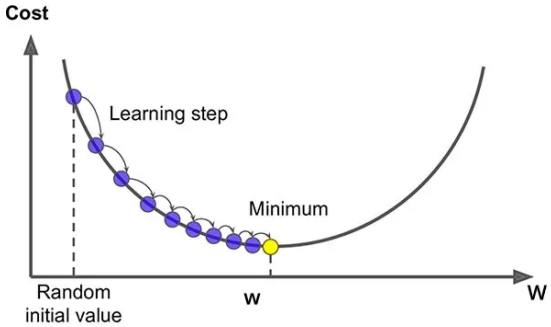
where Huber loss uses MSE logic for “small” residual values and MAE logic for “large” residual values.



Learning / Estimation / Fitting



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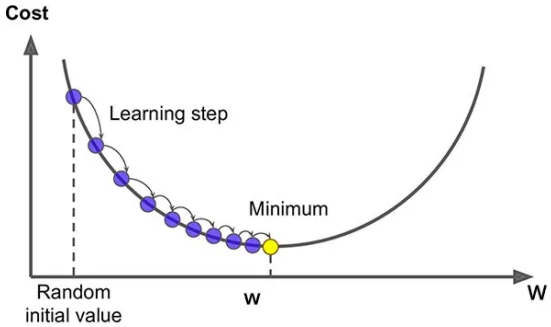

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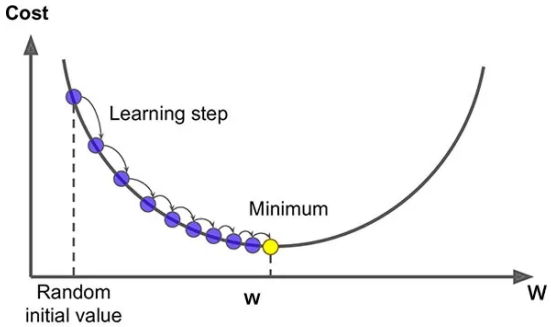

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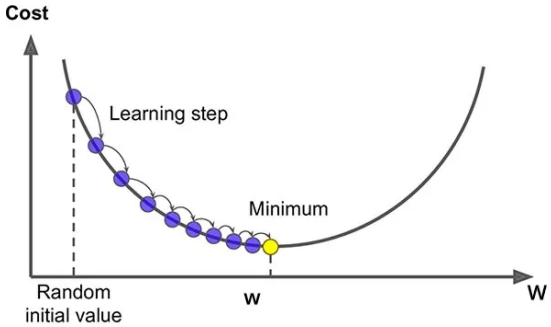

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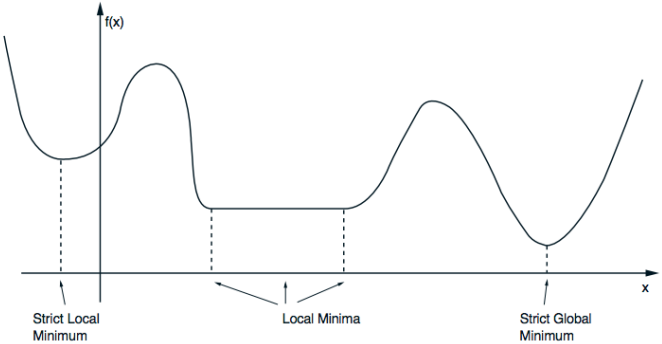
We will use an optimization algorithm like GD to solve the problem (to find a good $\mathbf{w}$).

Optimization landscapes



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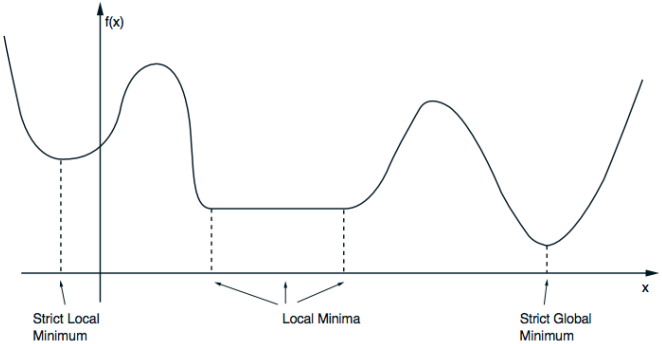

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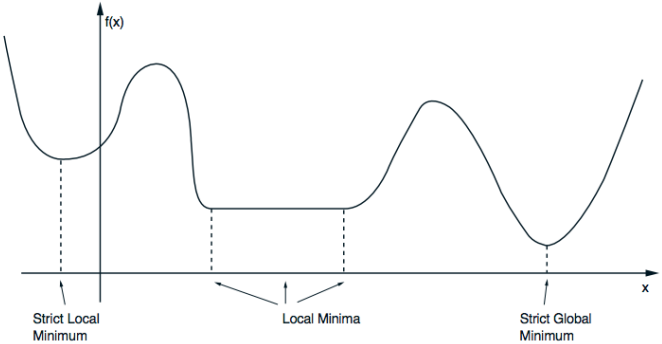

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$$f(x,y) = 3(1-x)^2 e^{-x^2-(y+1)^2} - 10(\frac{x}{5} - x^3 - y^5)e^{-x^2-y^2} - \frac{1}{3}e^{-(x+1)^2-y^2} \tag{12}$$

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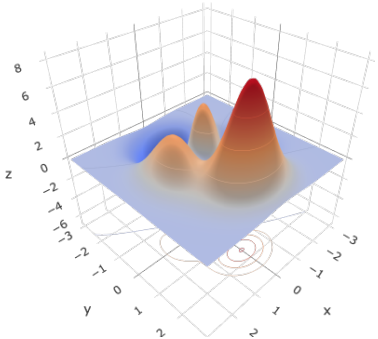

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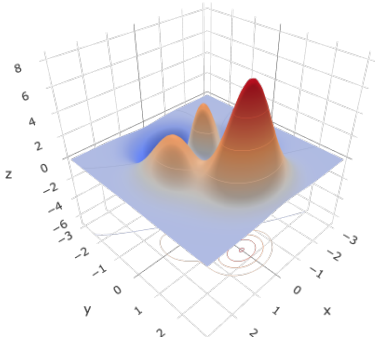

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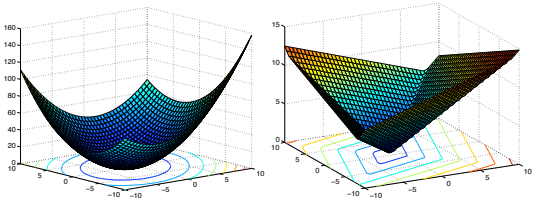
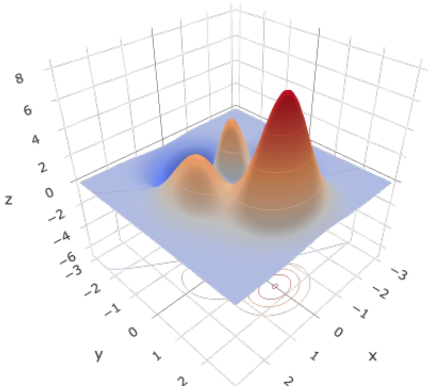
For a 2-parameter model, MSE$(\mathbf{w})$ and MAE$(\mathbf{w})$ are shown below.

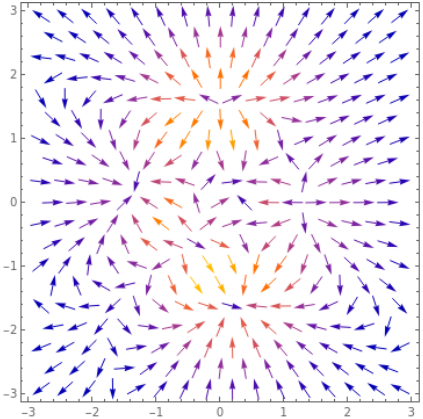
(We used $\mathbf{y}_n \approx w_0 + w_1 x_{n1}$ with $\mathbf{y}^\top = [2, -1, 1.5]$ and $\mathbf{x}^\top = [-1, 1, -1]$).



Example:



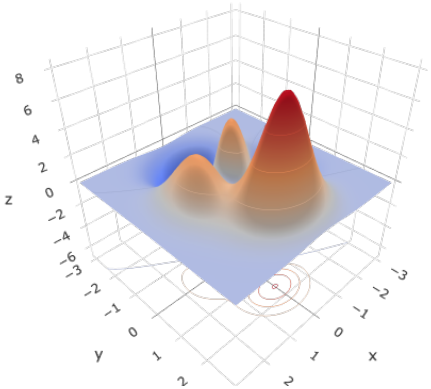
Cost function $f(x, y)$.



Gradient of f plotted as a vector field:

$$\nabla f(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x}(x, y) \\ \frac{\partial f}{\partial y}(x, y) \end{bmatrix}$$

Example: revisit this link for more details.



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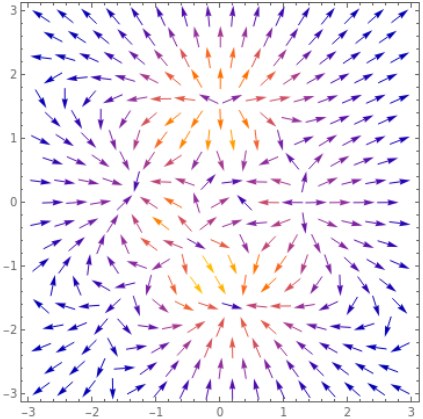

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where $\gamma > 0$ is the step-size (or learning rate). Then repeat with the next t.

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- Regression
- Classification

2 Linear Regression
- Definition of Linear Regression
- Optimization Basics
- Gradient Descent (GD) variants for Linear Regression
- Normal Equations and Least Squares

Gradient Descent for linear regression with MSE

Considering a dataset $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$ and learnable weights $\mathbf{w} \in \mathbb{R}^D$ for $f_{\mathbf{w}}(\mathbf{X}) = \mathbf{X}\mathbf{w}$.

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and then the gradient is given by $\nabla \mathcal{L}(\mathbf{w}) = -\frac{1}{N} \mathbf{X}^\top \mathbf{e}$.

Stochastic Gradient Descent (SGD)

- **Sum Objectives:** In machine learning, most cost functions are formulated as a sum over the training examples, that is

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- **Theoretical Motivation:** In expectation over the random choice of n, we have

$$\mathbb{E}[\nabla \mathcal{L}_n(\mathbf{w})] = \nabla \mathcal{L}(\mathbf{w})$$

  which is the true gradient direction.

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- Note that in the extreme case $B := [N]$, we obtain (batch) gradient descent, i.e. $\mathbf{g} = \nabla \mathcal{L}$.

An alternative view of SGD

Finite-sum empirical risk minimization problem:

$$f(\mathbf{x}^\star) = \min_{\mathbf{x}\in\mathbb{R}^d} \left\{ f(\mathbf{x}) := \frac{1}{N}\sum_{i=1}^{N} \left(f_i(\mathbf{x}) := F(\mathbf{x}, \boldsymbol{\xi}_i) \right) \right\}$$

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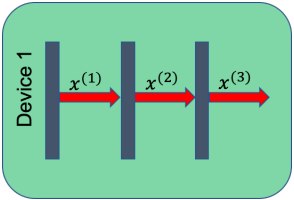

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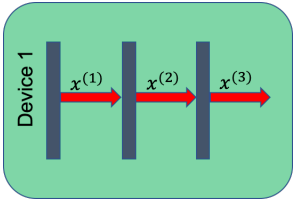

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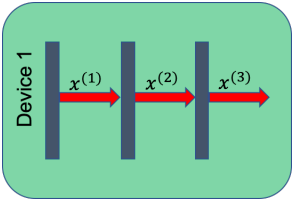

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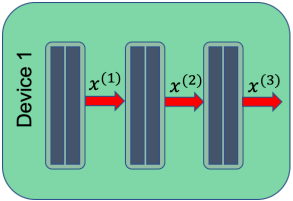

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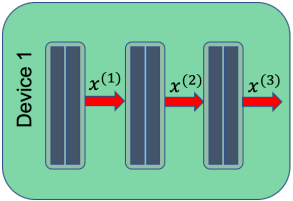
- $\mathcal{B} \in \{1, \ldots, N\}$ with $|\mathcal{B}| = B$.



Alternative view: Stochastic reformulation of finite-sum problems (SGD with arbitrary sampling)

Random sampling vector $\mathbf{v} = (v_1, \ldots, v_N) \sim \mathcal{D}$ with $\mathbb{E}[v_i] = 1$ for $i = 1, \ldots, N$.

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Original Finite-sum problem

$$\min_{\mathbf{x}\in\mathbb{R}^d} \frac{1}{N}\sum_{i=1}^N f_i(\mathbf{x})$$

$\longleftrightarrow$

Stochastic Reformulation

$$\min_{\mathbf{x}\in\mathbb{R}^d} \mathbb{E}[f_v(\mathbf{x})]$$

Minimizing the expectation of **random linear combinations** of original function

Alternative view: Stochastic reformulation of finite-sum problems (SGD with arbitrary sampling)

Random sampling vector $\mathbf{v} = (v_1, \ldots, v_N) \sim \mathcal{D}$ with $\mathbb{E}[v_i] = 1$ for $i = 1, \ldots, N$.

$$f(\mathbf{x}) := \tfrac{1}{N}\sum_{i=1}^N f_i(\mathbf{x}) = \tfrac{1}{N}\sum_{i=1}^N \mathbb{E}[v_i] f_i(\mathbf{x}) = \mathbb{E}\left[\underbrace{\frac{1}{N}\sum_{i=1}^N v_i f_i(\mathbf{x})}_{=: f_v(\mathbf{x})}\right] \qquad \text{(Stochastic Reformulation)}$$

Original Finite-sum problem

$$\min_{\mathbf{x}\in\mathbb{R}^d} \frac{1}{N}\sum_{i=1}^N f_i(\mathbf{x})$$

Sample $v_i^{(t)} \sim \mathcal{D}$

$$\mathbf{x}^{(t+1)} \leftarrow \mathbf{x}^{(t)} - \eta \nabla f_{v_i^{(t)}}(\mathbf{x}^{(t)})$$

Stochastic Reformulation

$$\min_{\mathbf{x}\in\mathbb{R}^d} \mathbb{E}[f_v(\mathbf{x})]$$

Minimizing the expectation of **random linear combinations** of original function

The distribution $\mathcal{D}$ encodes any form of mini-batching / non-uniform sampling.

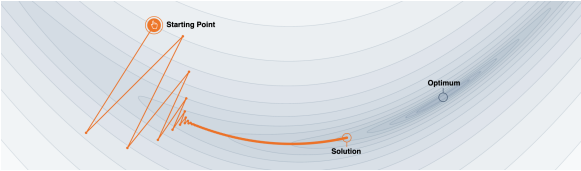
Table of Contents

1 Regression and Classification
- Regression
- Classification

2 Linear Regression
- Definition of Linear Regression
- Optimization Basics
- Gradient Descent (GD) variants for Linear Regression
- Normal Equations and Least Squares

Motivation

Recall that *GD for Linear Regression with MSE loss* is an <u>iterative</u> optimization algorithm, computationally flexibility for handling large datasets.



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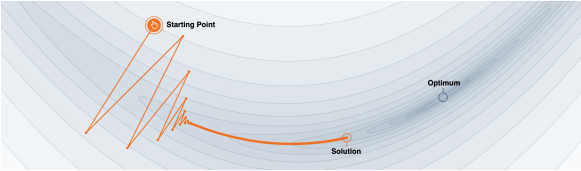
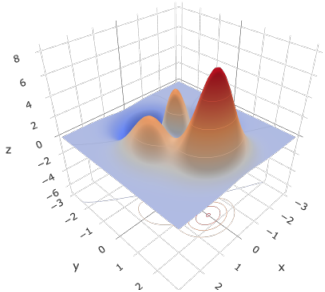
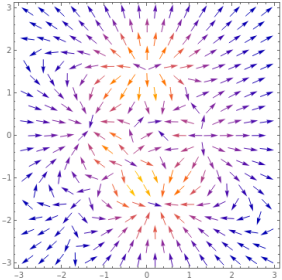


Can we compute the optimum of the cost function analytically?

- Linear regression using an MSE cost function is one such case.



Cost function $f(x, y)$.



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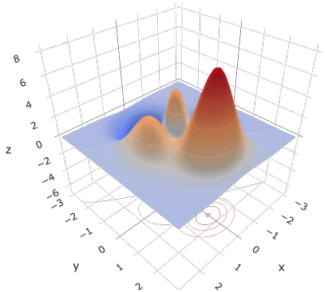

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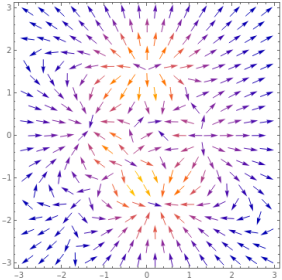

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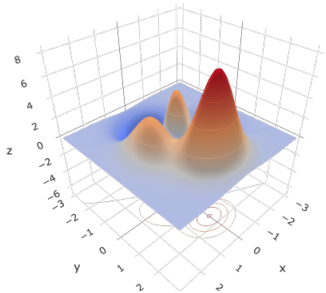

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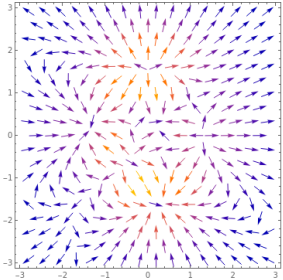

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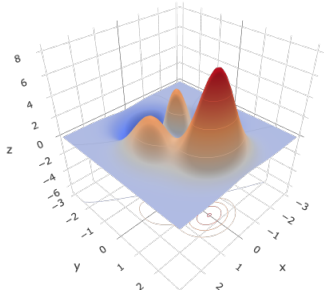

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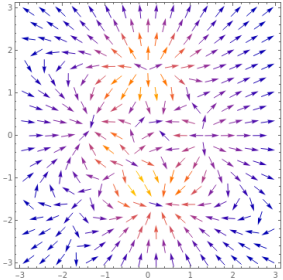

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where

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What is the meaning of finding the analytical solution?

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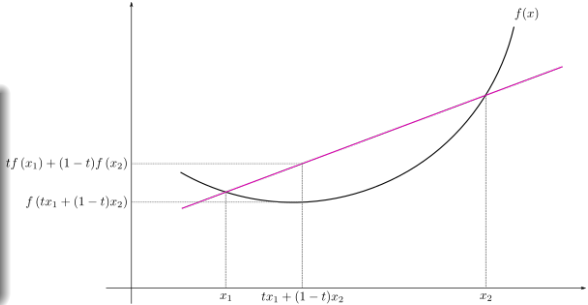

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A function $h(\mathbf{u})$ with $\mathbf{u} \in \mathbb{R}^D$ is convex, if for any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^D$ and for any $0 \leq \lambda \leq 1$, we have:

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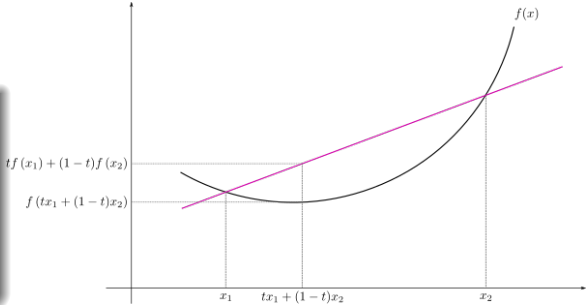

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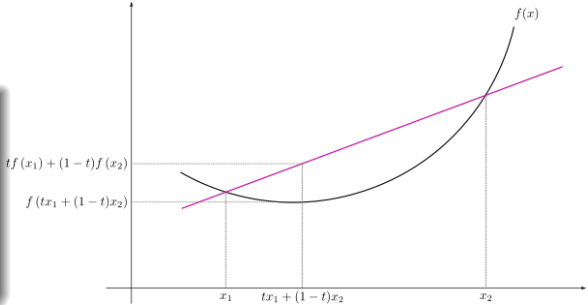

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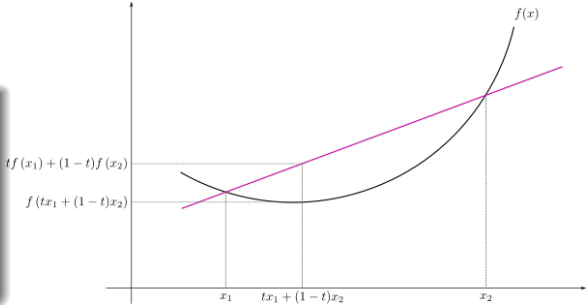

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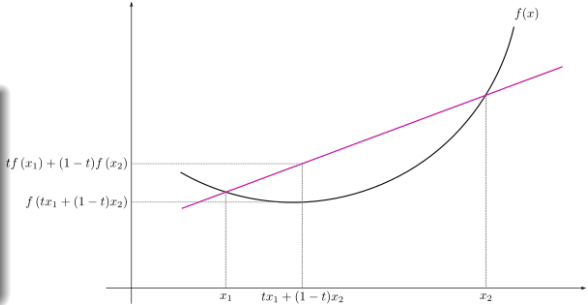

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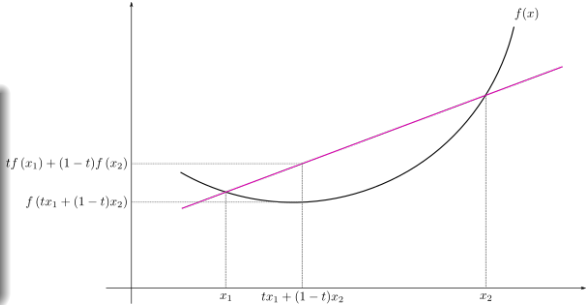

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$$\nabla \mathcal{L}(\mathbf{w}^\star) = \mathbf{0}\,, \tag{21}$$

where $\mathbf{w}^\star$ corresponds to the parameter at the optimum point.

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There are several ways of proving this:

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By taking the gradient of $\mathcal{L}(\mathbf{w}) = \frac{1}{2N}\sum_{n=1}^{N}\left(y_n - \mathbf{x}_n^\top \mathbf{w}\right)^2 = \frac{1}{2N}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top(\mathbf{y} - \mathbf{X}\mathbf{w})$, we have

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Derivation (step 2): finding the minimum of a convex function

By taking the gradient of $\mathcal{L}(\mathbf{w}) = \frac{1}{2N}\sum_{n=1}^{N}\left(y_n - \mathbf{x}_n^\top \mathbf{w}\right)^2 = \frac{1}{2N}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top(\mathbf{y} - \mathbf{X}\mathbf{w})$, we have

$$\nabla\mathcal{L}(\mathbf{w}) = -\frac{1}{N}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w})\,. \tag{24}$$

Given the property of convexity $\nabla\mathcal{L}(\mathbf{w}^\star) = \mathbf{0}$, we can get the normal equations for linear regression:

$$\mathbf{X}^\top \underbrace{(\mathbf{y} - \mathbf{X}\mathbf{w})}_{\text{error}} = \mathbf{0}\,, \tag{25}$$

where the error $\mathbf{e} := \mathbf{y} - \mathbf{X}\mathbf{w}$ is orthogonal to all columns of $\mathbf{X}$.

Geometric interpretation of Normal Equations

Definition 10 (Span of a set of vectors)

The **span** of a set of vectors, $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$, is the set of all possible linear combinations of these vectors; i.e. $\text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\} = \{\alpha_1\mathbf{x}_1 + \ldots + \alpha_k\mathbf{x}_k \mid \alpha_1, \ldots, \alpha_k \in \mathbb{R}\}$.

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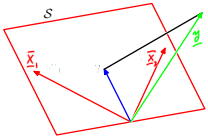

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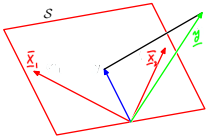

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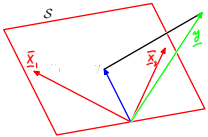

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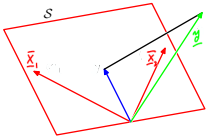

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- the optimum choice for $\mathbf{u}$, i.e. $\mathbf{u}^\star$, requires $\mathbf{y} - \mathbf{u}^\star$ to be orthogonal to $span(\mathbf{X})$.
- $\mathbf{u}^\star$ should be equal to *the projection of* $\mathbf{y}$ *onto* $span(\mathbf{X})$.

Least Squares

We need to solve the linear system of the normal equation $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w}) = \mathbf{0}$, where

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Remark 11

The Gram matrix $\mathbf{X}^\top\mathbf{X} \in \mathbb{R}^{D\times D}$ is invertible if and only if $\mathbf{X}$ has full column rank, or in other words $rank(\mathbf{X}) = D$.

Rank deficiency and ill-conditioning

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Can we solve least squares if $\mathbf{X}$ is rank deficient?
Yes, using a linear system solver, e.g., $\mathrm{np.linalg.solve}(\mathbf{X}, \mathbf{y})$.

This lecture:

- Basic concept of regression and classification
- Linear regression
  - Definition
  - Gradient Descent (GD) optimization
- Normal Equation and Least Squares

Next lecture:

- The probabilistic interpretation of Linear Regression
- Maximum Likelihood Estimation (MLE)
- Over-fitting and Under-fitting
- Polynomial Regression, Ridge Regression, and Lasso